

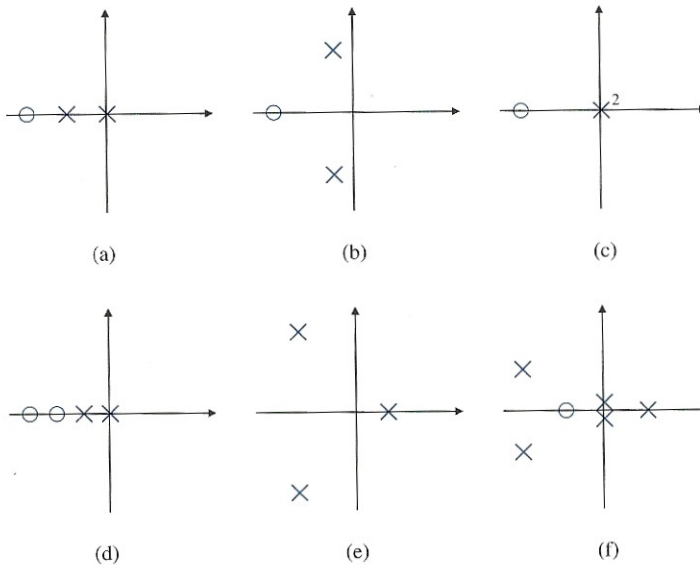
- (a) $s + (1/\tau) = 0$ versus parameter τ
- (b) $s^2 + cs + c + 1 = 0$ versus parameter c
- (c) $(s + c)^3 + A(Ts + 1) = 0$
- (i) versus parameter A ,
 - (ii) versus parameter T ,
 - (iii) versus the parameter c , if possible. Say why you can or cannot. Can a plot of the roots be drawn versus c for given constant values of A and T by any means at all?
- (d) $1 + \left[k_p + \frac{k_I}{s} + \frac{k_D s}{\tau s + 1} \right] G(s) = 0$. Assume that $G(s) = A \frac{c(s)}{d(s)}$, where $c(s)$ and $d(s)$ are monic polynomials with the degree of $d(s)$ greater than that of $c(s)$.
- (i) versus k_p
 - (ii) versus k_I
 - (iii) versus k_D
 - (iv) versus τ

Problems for Section 5.2: Guidelines for Sketching a Root Locus

- 5.2** Roughly sketch the root loci for the pole-zero maps as shown in Fig. 5.51 without the aid of a computer. Show your estimates of the center and angles of the asymptotes, a rough evaluation of arrival and departure angles for complex poles and zeros, and the loci for positive values of the parameter K . Each pole-zero map is from a characteristic equation of the form

$$1 + K \frac{b(s)}{a(s)} = 0,$$

where the roots of the numerator $b(s)$ are shown as small circles $\circ$ and the roots of the denominator $a(s)$ are shown as $\times$'s on the s -plane. Note that in Fig. 5.51(c) there are two poles at the origin.



Problems for Section 5.5: A Design Example Using the Root Locus

5.35 Consider the positioning servomechanism system shown in Fig. 5.65, where

$$e_i = K_{pot}\theta_i, \quad e_o = K_{pot}\theta_o, \quad K_{pot} = 10 \text{ V/rad},$$

$$T = \text{motor torque} = K_t i_a,$$

$$k_m = K_t = \text{torque constant} = 0.1 \text{ N-m/A},$$

$$R_a = \text{armature resistance} = 10 \Omega,$$

$$\text{Gear ratio} = 1 : 1,$$

$$J_L + J_m = \text{total inertia} = 10^{-3} \text{ kg-m}^2,$$

$$C = 200 \mu\text{F},$$

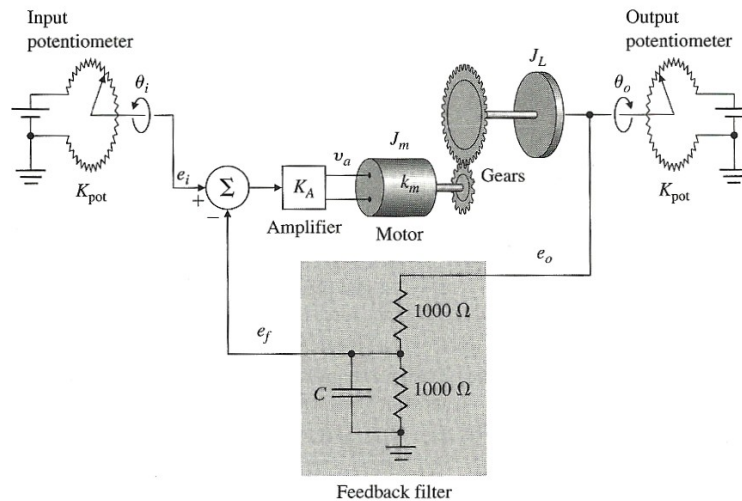
$$v_a = K_A(e_i - e_f).$$

- (a) What is the range of the amplifier gain K_A for which the system is stable? Estimate the upper limit graphically using a root-locus plot.
- (b) Choose a gain K_A that gives roots at $\zeta = 0.7$. Where are all three closed-loop root locations for this value of K_A ?

Figure 5.65

Positioning servomechanism

Source: Reprinted from Clark, 1982, with permission



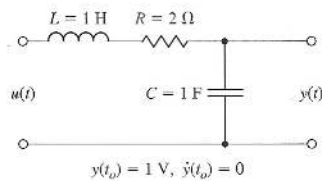
- 5.36 We wish to design a velocity control for a tape-drive servomechanism. The transfer function from current $I(s)$ to tape velocity $\Omega(s)$ (in millimeters per millisecond per ampere) is

$$\frac{\Omega(s)}{I(s)} = \frac{15(s^2 + 0.9s + 0.8)}{(s + 1)(s^2 + 1.1s + 1)}.$$

We wish to design a Type 1 feedback system so that the response to a reference step satisfies

$$t_r \leq 4 \text{ msec}, \quad t_s \leq 15 \text{ msec}, \quad M_p \leq 0.05.$$

Figure 7.82
Circuit for Problem 7.1



initial conditions shown in the figure. Verify your answer using the initial command in MATLAB.

- 7.2 A schematic for the satellite and scientific probe for the Gravity Probe-B (GP-B) experiment that was launched on April 30, 2004 is sketched in Fig. 7.83. Assume that the mass of the spacecraft plus helium tank, m_1 , is 2000 kg and the mass of the probe, m_2 , is 1000 kg. A rotor will float inside the probe and will be forced to follow the probe with a capacitive forcing mechanism. The spring constant of the coupling k is 3.2×10^6 . The viscous damping b is 4.6×10^3 .

- Write the equations of motion for the system consisting of masses m_1 and m_2 using the inertial position variables, y_1 and y_2 .
- The actual disturbance u is a micrometeorite, and the resulting motion is very small. Therefore, rewrite your equations with the scaled variables $z_1 = 10^6 y_1$, $z_2 = 10^6 y_2$, and $v = 1000u$.
- Put the equations in state-variable form using the state $\mathbf{x} = [z_1 \ \dot{z}_1 \ z_2 \ \dot{z}_2]^T$, the output $y = z_2$, and the input an impulse, $u = 10^{-3} \delta(t)$ N-sec on mass m_1 .
- Using the numerical values, enter the equations of motion into MATLAB in the form

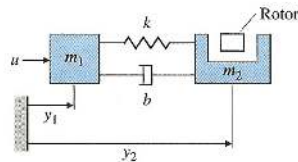
$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}v, \quad (7.261)$$

$$y = \mathbf{H}\mathbf{x} + \mathbf{J}v. \quad (7.262)$$

and define the MATLAB system: `sysGPB = ss(F,G,H,J)`. Plot the response of y caused by the impulse with the MATLAB command `impz(sysGPB)`. This is the signal the rotor must follow.

- Use the MATLAB commands `p = eig(F)` to find the poles (or roots) of the system and `z = tzero(F,G,H,J)` to find the zeros of the system.

Figure 7.83
Schematic diagram of
the GP-B satellite and
probe



Problems for Section 7.4: Analysis of the State Equations

- 7.3 Give the state description matrices in control-canonical form for the following transfer functions:

- $\frac{1}{4s+1}$